databricks

Catch OEE losses before they happen

IT + OT on one Databricks platform · turbine-component plant

FEBAR build · Narasimhan Raam, Sr. Solutions Architect

The outcome, per line, per year

One 12-asset line, base case (assumptions on slide 8)

- +1.0 OEE point (range +0.7 to +1.7) by fixing degradation before it becomes a breakdown
- ~500 fewer unplanned downtime hours and ~2,500 fewer scrapped parts
- ≈ $2.75M a year (range $1.0M to $7.7M), before secondary-damage and throughput upside
- Failures flagged 57–86 h ahead; the model beats today's rule (precision 0.82 vs 0.67)
- Supervisor acted 2.1 min after the alert, with procedure, parts and work order ready



Who it is for and what each gets

Exec sponsor

- VP Manufacturing Operations
- Plant OEE toward the 75 % target without capital spend
- Fewer missed shipments from unplanned stops
- One governed OEE number across plants

Domain owner

- Plant Reliability & Maintenance Engineering Manager
- Next-shift risk per asset, with the reason
- AI-drafted work orders tied to the manual procedure
- Run-to-failure turned into planned work

Technical owner

- IT/OT Architect
- Historian/SCADA and ERP/MES/CMMS on one asset key
- One platform to ingest, govern, serve, predict, explain
- Least privilege and lineage by default



Where OEE is lost: the hidden 13 %

28-day baseline · 12 assets · 1,008 asset-shifts	All other shifts	4 days before a breakdown (133 shifts)
OEE	76.5 %	58.4 %
Availability	90.6 %	86.2 %
Performance (speed loss, minor stops)	86.2 %	73.0 %
Quality (scrap rate)	97.9 % (1.9 % scrap)	92.6 % (6.9 % scrap)
Unplanned downtime	77 h	63 h, including all 10 breakdowns (5.3 h each)
Plant OEE overall	74.1 % against a 75 % target	Closing these windows alone is worth +2.4 points



One data journey, one asset key

Stage	Databricks capability	What it does here
1 Ingest	Lakeflow declarative pipeline + Auto Loader	1.9 M historian readings, SCADA states/alarms, ERP/MES orders and CMMS work orders; 18 data-quality expectations drop bad sensor data, unmapped tags and impossible counts
2 Govern	Unity Catalog	IT / OT / gold domains with owner + steward tags, certified gold OEE, operator PII masked, lineage from raw files to OEE
3 Predict + explain	MLflow model in UC + ai_query	Next-shift OEE-at-risk model with SHAP drivers; LLM drafts root cause + work order grounded on past repairs
4 Serve + act	Lakebase (Postgres)	Sub-millisecond at-risk queue for the app; supervisor actions written back and synced to the lakehouse
5 Ask	Genie + Agent Bricks Supervisor + Knowledge Assistant	Plain-English OEE analytics, risk, and repair manuals in one agent; evaluated with MLflow LLM judges
6 Use	Databricks App	OEE board, at-risk queue, Acknowledge / Create work order, agent chat with human approval for actions



On the floor: CNC-02 before failure

What the supervisor sees and does

- Signal: vibration 6.0 mm/s, +1.8 vs the last 6 shifts; 8 alarms; OEE down to 52 %
- Prediction: 99.8 % risk of next-shift OEE < 65 %; top driver is the vibration trend
- Cause: likely spindle-bearing wear from lubrication starvation (4 similar past repairs)
- Plan: procedure MM-CNC-01-R7 (6 h), pre-stage bearing set SP-BRG-7014-P4, P1 at next planned stop
- Action: agent raises the request only after the user approves; supervisor approves the work order
- Loop closed in Unity Catalog: first action 2.1 min after the alert

Proof: model and agent, measured

Held-out last 8 days (276 asset-shifts)	At-risk model	Today's rule: flag if this shift's OEE < 65 %
Precision (alerts that were right)	0.82	0.67
Recall (at-risk shifts caught)	0.72	0.65
AUC	0.87	—
Warning before the 3 breakdowns in the window	57–86 h	after OEE has already dropped
Agent eval: 10 plant questions, MLflow judges	Routing 1.0 · relevance 1.0 · safety 1.0	Plant rules 1.0 · correctness 0.8



Business value per 12-asset line, per year

Lever	Low	Base	High	Key assumption (to replace with plant data)
OEE points	+0.7	+1.0	+1.7	72 % of windows caught; 60 % of each window's loss recovered
Unplanned hours avoided	350	500	500	breakdown hours × detection rate
Downtime value	$0.6 M	$1.75 M	$3.5 M	$2.5k–$10k per unplanned hour on a constrained cell; planned hour = 30 %
Scrap avoided (parts)	1,700	2,500	4,200	excess scrap in degradation windows
Scrap value	$0.35 M	$1.0 M	$4.2 M	$200–$1,000 per in-process blade / vane part
Total	≈ $1.0 M	≈ $2.75 M	≈ $7.7 M	excludes secondary damage, throughput, MTTR gains



Trust by design

Governed data

- IT, OT and gold domains, each with an owner and a steward
- Operator names and badges masked outside the HR group
- Lineage from historian file to the OEE number
- App identity limited to the 7 tables, 2 functions, 2 endpoints and 1 Genie space it needs

Safe AI and actions

- Model registered with a signature; drivers shown per asset
- Answers grounded on plant data and manuals, procedure IDs cited
- Agent actions need two human approvals: tool call, then work order
- Every question and action logged and synced back for audit



Roadmap and the ask

Phase	Scope	Exit criterion
Days 0–30 · Connect	One line: real historian tags + SCADA states, ERP/MES orders, CMMS work orders; asset key map	Daily OEE per asset and shift matches the plant's report within ±1 point
Days 31–60 · Predict	Retrain on 6–12 months of history; add manuals; pilot app with one shift team	Precision ≥ 0.75 on held-out weeks; supervisors act on alerts within 1 hour (SOP)
Days 61–90 · Prove	Run in production on the line; track planned vs unplanned stops	Measured OEE and downtime change vs baseline; go / no-go for plant rollout
Ask	Sponsor: approve a 90-day pilot on one line. Domain owner: name a pilot shift team and the reliability engineer	Plant data access (historian, MES, CMMS) granted in week 1



Appendix: evidence map

Claim	Text evidence (GitHub repo)
Pipeline ran; quality rules caught the bad data	evidence/01, evidence/02 (event-log expectations)
Masking, tags, lineage	evidence/03
Model metrics, warning lead time, SHAP drivers	evidence/04, evidence/05
Lakebase serving, reverse sync, 0.02 ms lookup	evidence/06, evidence/07
Genie, Knowledge Assistant, Supervisor answers; LLM-judge scores	evidence/08, evidence/09
App grants, closed-loop action, 2.1 min KPI	evidence/10, evidence/06 (Postgres grants)
Value math and assumptions	docs/deck/VALUE_MODEL.md




databricks